

Module 4

Shell Productivity & Basic Scripting

Learning objectives

- Use history and Ctrl+R effectively
- Write scripts with shebang and chmod +x
- Pass arguments with \$1 and \$@
- Apply set -e and quoted variables

Prerequisites

- Module 3 (redirection, pipes)

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[History & aliases] --> B[Scripts]

B --> C[Variables]

C --> D[Control flow]

D --> E[Safe scripting]

E --> F[Module 5]

Course context

- Wrap compile/sim steps in scripts/
- Course module scripts are the pattern to follow
- Quote variables to avoid word-splitting bugs

Core topics

Self-check

```
$ ./scripts/module4.sh --check
```

Terminal: module_check

```
[[0;36m===== [[0m
[[0;36mModule 4: Environment self-check [[0m
[[0;36m===== [[0m

[[0;32m[OK] [[0m bash available
[[0;32m[OK] [[0m module4 directory exists: /home/yongfu/proj/universal-verification-methodology/
[[0;32m[OK] [[0m script_basics/say_hello.sh exists
[[0;32m[OK] [[0m arguments/greet.sh exists
[[0;32m[OK] [[0m control_flow/count_lines.sh exists
[[0;32m[OK] [[0m safe_scripting/clean_tmp.sh exists

[[0;32m[OK] [[0m All required checks passed.
```

Minimal script

```
#!/usr/bin/env bash  
echo "Hello from script"
```

First bash script

```
$ cd module4/examples/script_basics && chmod u+x say_hello.sh &&  
./say_hello.sh
```

```
Terminal: say_hello  
Hello from the script!
```

Positional parameters

```
$ cd module4/examples/arguments && chmod u+x greet.sh && ./greet.sh  
Student
```

Terminal: greet_args

Hello, Student!

Practice

Exercise scaffold

```
./scripts/module4.sh --scaffold
```

Exercises

- Run `say_hello.sh` and `greet.sh` from scaffold
- Add an alias `ll='ls -aF'`
- Write a loop over `*.log`

Summary & next steps

- Scripts automate design flows
- History and aliases save time
- Next: Module 5 — projects & archives
- Next: Module 5: Editors, Projects & Archives